Lutetium (^{177}Lu) oxodotreotide (Lutathera®) Therapy

INDICATIONS FOR USE:

INDICATION	ICD10*	Regimen Code	HSE approved reimbursement status**
Treatment of unresectable or metastatic, progressive, well differentiated (G1 and G2), somatostatin receptor positive-gastroenteropancreatic neuroendocrine tumours (GEP- NETs) in adults.	C16-20, C25, C26	00642a	ODMS 01/02/2021

*Neuroendocrine tumours can occur in a variety of anatomical locations, C16-20 & C25 refer to gastroenterohepatic tumours, where exact location cannot be determined C26 can be used.

** This applies to post 2012 indications.

TREATMENT:

Lutetium (^{177}Lu) oxodotreotide is administered IV, **once every 8 weeks for 4 infusions.** The infusion interval maybe extended up to 16 weeks in case of dose modifying toxicity (DMT) (See dose modifications).

It should be administered only by persons authorised to handle radiopharmaceuticals in designated clinical settings which satisfy radiation safety and regulation requirements.

Facilities to treat anaphylaxis MUST be present when lutetium (^{177}Lu) oxodotreotide is administered.

Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	Lutetium (^{177}Lu) oxodotreotide[a]	7,400 MBq	IV infusion	Over approximately 30 to 40 minutes[b]	Every 8 weeks[c]
[a]For renal protection, an intravenous amino acid solution must be administered intravenously over 4 hours, starting 30 minutes before infusion of lutetium (^{177}Lu) oxodotreotide. Please see pre-medication section for suggested amino acid solutions.					
[b]Lutetium (^{177}Lu) oxodotreotide infusion should start 30 minutes after the beginning of the amino acid solution infusion, with an infusion rate of up to 400 mL/h.Please refer to the SmPC for specific instructions for administration using gravametric, or peristaltic syringe pump method.					
[c]The infusion interval maybe extended up to 16 weeks in case of DMT.					

ELIGIBILITY:

- Indications as above
- Patients ≥18 years
- Well differentiated locally advanced or metastatic gastroenteropancreatic neuroendocrine tumours that had progressed on standard octreotide treatment
- Previous treatment with long acting somatostatin analogue at a fixed dose for at least 12 weeks prior

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

- Confirmation of tumour somatostatin receptor overexpression by imaging (scintigraphy or positron emission tomography [PET])
- ECOG 0-2
- Adequate renal, hepatobiliary, cardiac and haematological function

CAUTION:

- Given the mechanism of action and the tolerance profile of lutetium (^{177}Lu) oxodotreotide, caution is recommended in the following cases:
    - Previous external beam radiotherapy involving more than 25% of the bone marrow
    - Severe heart failure defined as class III or IV in the New York Heart Association (NYHA) classification
    - Kidney failure with creatinine clearance < 40 mL/min
    - Impaired haematological function with either Hb < 8 g/dL, platelets < 75 x 10^9/L or leucocytes < 2 x 10^9/L (except lymphopenia)
    - Liver impairment with either total bilirubin > 3 times the upper limit of normal (ULN), regardless of AST level or albuminaemia <30g/L and INR >1.5

EXCLUSIONS:

- Hypersensitivity to lutetium (^{177}Lu) oxodotreotide or any of the excipients.
- Patients who are candidates for curativesurgery
- CrCl<30mL/minute
- Pregnancy and breastfeeding

PRESCRIPTIVE AUTHORITY:

Therapy Plan:

The therapy plan must be determined by the patient's primary consultant following discussion and agreement with the patient and their carer(s) if required. The primary consultant for this indication is a consultant with specialist interest in the management and treatment of NETs* where the treatment is to be administered in RPII licensed sites where appropriate supporting infrastructure is in place. Consideration should also be given to the broader care and management of patients associated with the use of radionuclides including:

1. The recommendation to treat as agreed at the relevant MDM for NETs.
2. The prescription, administration and oversight of the radionuclide treatment itself.
3. Follow up during/after treatment to ensure safety and to evaluate treatment response and the patient's capacity to continue with the next course of treatment.

*This may be the nuclear medicine consultant/ medical oncologist or other consultant with expertise in the management and treatment of NET with radionuclides.

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Cardiac function (ECHO or MUGA)
- Pregnancy test

Regular tests:

- FBC, renal and liver profile at least once in the 2-4 weeks prior to administration, and shortly before administration. These tests should also be performed every 2 weeks between treatments, every 4 weeks for at least 3 months after the last infusion of lutetium (^{177}Lu) oxodotreotide, and every 3 months for the following year. Renal function as determined by serum creatinine and calculated creatinine clearance must be assessed at baseline, during and at least for the first year after treatment

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.
- Management of severe or intolerable adverse drug reactions may require temporary dose interruption, extension of the dosing interval from 8 weeks up to 16 weeks, dose reduction, or permanent discontinuation of treatment with lutetium (^{177}Lu) oxodotreotide

Haematological:

Table 3: Dose modification lutetium (^{177}Lu) oxodotreotide for haematological toxicity

Toxicity	Severity		Dose Modification
Thrombocytopenia	**Grade**	**Platelets (x10^9/L)**	Withhold dose until complete or partial resolution (Grade 0 to 1). Resume treatment at 3,700 MBq (100 mCi) in patients with complete or partial resolution. If reduced dose does not result in Grade 2, 3 or 4 thrombocytopenia, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for Grade ≥2 thrombocytopenia requiring a treatment delay of 16 weeks or longer.
	2	50-75[a]	
	3	25-49	
	4	<25	
	Recurrent Grade 2, 3 or 4		Permanently discontinue treatment.
Neutropenia	**Grade**	**ANC (x10^9/L)**	
	3	0.5 – 0.99	Withhold dose until complete or partial resolution (Grade 0, 1, or 2). Resume treatment at 3,700 MBq (100 mCi) in patients with complete or
	4	<0.5	

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

			partial resolution. If reduced dose does not result in Grade 3 or 4 neutropenia, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for Grade ≥3 neutropenia requiring a treatment delay of 16 weeks or longer.
	Recurrent grade 3 or 4		Permanently discontinue treatment.
Anaemia	**Grade**	**Hgb (g/dL)**	
	3	<8.0[a]	Withhold dose until complete or partial resolution (Grade 0, 1, or 2). Resume treatment at 3,700 MBq (100 mCi) in patients with complete or partial resolution. If reduced dose does not result in Grade 3 or 4 anaemia, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for Grade ≥3 anaemia requiring a treatment delay of 16 weeks or longer.
	4	Life threatening consequences	
	Recurrent Grade 3 or 4		Permanently discontinue treatment.
[a]The same thresholds are also applicable to baseline values at the time of treatment initiation			

Renal and Hepatic Impairment:

Careful consideration to the administration of lutetium (^{177}Lu) oxodotreotide is required since an increased radiation exposure is possible in patients with renal and hepatic impairment.

Table 4: Dose modification of lutetium (^{177}Lu) oxodotreotide in renal and hepatic impairment

Renal Impairment	Hepatic Impairment
The pharmacokinetic profile and safety of lutetium (^{177}Lu) oxodotreotide in patients with severe renal impairment or end-stage renal disease have not been studied. Treatment with lutetium (^{177}Lu) oxodotreotide in patients with severe kidney failure (CrCl <30 mL/min) is contraindicated. Treatment with lutetium (^{177}Lu) oxodotreotide in patients with CrCl <40 mL/min at baseline is not recommended. No dose adjustment is recommended for renally impaired patients with CrCl ≥40 mL/min. Renal function should be more frequently monitored during treatment as these patients may be at greater risk of toxicity. For additional details about the treatment of patients with renal toxicity, see Table 5	No dose adjustment is recommended for patients with mild or moderate hepatic impairment The pharmacokinetic profile and safety of lutetium (^{177}Lu) oxodotreotide in patients with severe hepatic impairment (bilirubin >3x ULN, regardless of AST) have not been studied therefore these patients should only be treated with lutetium (^{177}Lu) oxodotreotide after careful benefit-risk assessment. For additional details about the treatment of patients with mild to moderate hepatic toxicity, please see Table 5.

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

Table 5: Dose Modification of Lutetium (^{177}Lu) oxodotreotide for non-haematologic toxicity

Toxicity	Severity	Dose Modification
Renal toxicity	Defined as: • CrCl < 40 mL/min[a,b] or • 40% increase in baseline serum creatinine or • 40% decrease in baseline CrCl[b]	Withhold dose until complete resolution or return to baseline. Resume treatment at 3,700 MBq (100 mCi) in patients with complete resolution or return to baseline. If reduced dose does not result in renal toxicity, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for renal toxicity requiring a treatment delay of 16 weeks or longer
	Recurrent renal toxicity	Permanently discontinue treatment
Hepatotoxicity	Defined as: • Bilirubinaemia > 3 times the upper limit of normal (Grade 3 or 4)[c] or • Albuminaemia[c] less than 30 g/L with international normalised ratio (INR) >1.5	Withhold dose until complete resolution or return to baseline. Resume treatment at 3,700 MBq (100 mCi) in patients with complete resolution or return to baseline. If reduced treatment dose does not result in hepatotoxicity, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for hepatotoxicity requiring a treatment delay of 16 weeks or longer.
	Recurrent hepatotoxicity	Permanently discontinue treatment
Any other CTCAE* Grade 3 or Grade 4 toxicity	Grade 3 or 4	Withhold dose until complete or partial resolution (Grade 0 to 2). Resume treatment at 3,700 MBq (100 mCi) in patients with complete or partial resolution. If reduced dose does not result in Grade 3 or 4 toxicity, administer treatment at 7,400 MBq (200 mCi) for next dose. Permanently discontinue treatment for Grade 3 or higher toxicity requiring treatment delay of 16 weeks or longer.
	Recurrent Grade 3 or 4	Permanently discontinue treatment
[a]The same thresholds are also applicable to baseline values at the time of treatment initiation		
[b]calculate using Cockcroft-Gault with actual body weight		
[c]If the same thresholds are seen at baseline, treatment initiation to be considered after benefit risk assessment		
*CTCAE: Common Terminology Criteria for Adverse Events, National Cancer Institute		

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting- Available on the NCCP website

Lutetium (^{177}Lu) oxodotreotide: Moderate **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

PREMEDICATIONS:

- Premedication with 5-HT_3 antagonist should be used to help prevent nausea/vomiting associated with the amino acid infusion, should be commenced at least 30 minutes prior to the start of amino acid solution infusion to reach the full antiemetic efficacy.
  Depending on the amino acid solution, additional antiemetics including NK1 inhibitors and H_2 receptor antagonists may be required.
  Note: Corticosteroids should not be used as first-line anti-emetics (see drug interactions section below) but may be used in cases of refractory nausea/vomiting.
- For renal protection, an intravenous amino acid solution must be administered intravenously over 4 hours, starting 30 minutes before infusion of lutetium (^{177}Lu) oxodotreotide. The composition of the standard amino acid solution is detailed in Table 6. Alternatively, some commercially available amino acid solutions can be used if compliant with the specification described in Table 7 below.
- An amino acid solution containing just lysine and arginine in the amounts specified in Table 6 is considered the medicinal product of choice, due to the lower total volume to be infused and lower osmolality.

Table 6: Composition of the standard amino acid solution

Compound	Amount
L-Lysine HCl	25g*
L-Arginine HCl	25g**
NaCl 0.9% solution for injection, or water for injection	1L
*equivalent to 20g lysine **equivalent to 20.7g arginine	

Table 7. Specification of commercially available amino acid solutions

Characteristic	Specification
L-Lysine HCl content	Between 18 and 25g*
L-Arginine HCl content	Between 18 and 25g**
Volume	1L to 2L
Osmolality	<1,200 mOsmol/kg
*equivalent to 14.4 to 20g lysine **equivalent to 14.9 to 20.7g arginine	

OTHER SUPPORTIVE CARE:

- Concomitant use of somatostatin analogues may be needed for disease symptom control. Administration of long acting somatostatin analogues (SSAs) should be avoided within 30 days prior to the administration of lutetium (^{177}Lu) oxodotreotide. If necessary to control carcinoid syndrome, patients may be treated with short acting somatostatin analogues during the 4 weeks preceding lutetium (^{177}Lu) oxodotreotide administration, until 24 hours before the administration of lutetium (^{177}Lu) oxodotreotide. Long acting SSA can be administered 24 hours after administration of lutetium (^{177}Lu) oxodotreotide, can also be given on the same evening following luterium (177Lu) oxodotreotide if patient is treated as a day case
- Local Radiation Protection rules should be followed when administering lutetium (^{177}Lu) oxodotreotide

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens

- In patients with functioning NETs, particularly with high tumour burden, administration of lutetium (^{177}Lu) oxodotreotide may precipitate carcinoid crisis or other neuroendocrine hormonal crisis. Treating clinicians should be aware of this risk and local guidelines for carcinoid crisis management should be followed

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for full details.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

COMPANY SUPPORT RESOURCES/Useful Links:

Please note that this is for information only and does not constitute endorsement by the NCCP

Lutathera® Treatment Procedure Guide: https://assets.hpra.ie/products/Human/10514/9209fa2a-4261-4b39-802b-b98f837ce73c.pdf

REFERENCES:

1. Strosberg J, et al. Phase 3 Trial of 177Lu-Dotatate for Midgut Neuroendocrine Tumors. N Engl J Med. 2017 Jan 12; 376 (2):125-135. doi: 10.1056/NEJMoa1607427. PubMed PMID: 28076709; PubMed Central PMCID: PMC5895095. (Including supplementary material)
2. Strosberg |J et al. Health-Related Quality of Life in Patients With Progressive Midgut Neuroendocrine Tumours Treated With 177Lu-Dotatate in the Phase III NETTER-1 Trial. Journal of Clinical Oncology 2018. 36-25) 2578-2584
3. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
4. Lutetium (^{177}Lu) oxodotreotide (Lutathera®) Summary of Product characteristics. Accessed March 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/lutathera-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	19/09/2023		Dr Nicola Hughes and Dr Mark Doherty
1a	28/05/2024	Updated ICD-10 code	NCCP
2	27/05/2025	Regimen reviewed. Updated treatment table footnotes. Updated cautions section. Updated baseline testing section. Updated regimen in line with NCCP standardisation.	Dr Nicola Hughes

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

The information contained in this document is a statement of consensus of NCCP and ISMO or IHS professionals regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult these documents is expected to use independent medical judgement in the context of individual clinical circumstances to determine any patient's care or treatment. Use of these documents is the responsibility of the prescribing clinician and is subject to HSE's terms of use available at http://www.hse.ie/eng/Disclaimer

This information is valid only on the day of printing, for any updates please check www.hse.ie/NCCPSACTregimens